

Correlation-Aware Byzantine Fault Tolerance

Benjamin Haag¹ and Christian Berger¹

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1 Motivation

Typical BFT systems, such as BFT-SMaRt, tolerate up to f Byzantine replicas in a system of $n = 3f + 1$ replicas [BSA14]. As the system grows, reaching this fault threshold becomes increasingly unlikely if replica failures occur independently. However, replicas in real-world deployments often share dependencies, such as their geographic region, operator, or software stack. These dependencies form *failure domains* and can cause replica failures to become correlated [Me25; So23]. This becomes particularly problematic when more than f replicas belong to the same failure domain: a single correlated failure may then exceed the assumed fault threshold and compromise the entire system. Simply adding more replicas to the system does not address this issue. Instead, correlations between replicas must be explicitly captured by the fault model and considered by the quorum system to improve resilience against correlated failures. In this talk, we present Correlation-Aware Byzantine Fault Tolerance (CaBFT), an approach that explicitly incorporates such failure domains into the fault model and quorum construction of a BFT system.

2 Model

Let $R = \{r_1, \dots, r_n\}$ denote the set of replicas. Every replica $r_i \in R$ is associated with an attribute vector describing its failure domains.

$$r_i = (\text{zone}_i, \text{operator}_i, \text{software}_i) \quad (1)$$

Replicas with the same attribute value in a given dimension form a failure domain. We denote the set of all failure domains by $\mathcal{D}$. A correlated failure occurs when multiple replicas within such a failure domain fail due to a common cause. To tolerate the failure of an entire domain, no failure domain may contain more than f replicas. For $n = 3f + 1$, this corresponds to less than one third of all replicas. We refer to this property as the *Three-Thirds Diversity Requirement (TTDR)*.

¹ Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany, benny.haag@fau.de; berger@cs.fau.de

Real-world deployments do not necessarily satisfy TTDR, and enforcing it may be impractical. Instead, we generalize the requirement to replica weights. The resulting *Weighted Three-Thirds Diversity Requirement (WTTDR)* requires the accumulated weight of each failure domain to remain below one third of the total voting weight.

$$\forall D \in \mathcal{D} : \sum_{r_i \in D} w_i < \frac{1}{3} \sum_{r_i \in R} w_i \quad (2)$$

Based on WTTDR and the requirements for Byzantine quorum systems [MR98], we derive a weighted quorum system that can tolerate correlated failures even when individual failure domains are overrepresented by replica count.

3 CaBFT

We implemented CaBFT as an extension of BFT-SMaRt [BSA14], integrating our correlation-aware weighted quorum system into a practical BFT protocol. CaBFT uses the attribute vectors of all participating replicas to determine a suitable weight distribution by formulating the weight assignment as a mixed-integer programming (MIP) problem. The protocol supports dynamic reconfiguration through replica join and leave operations and reoptimizes the weight distribution whenever the replica configuration changes. Furthermore, leader selection is correlation-aware: previous leader failures are attributed to their respective failure domains and influence the selection of subsequent leaders.

While CaBFT demonstrates how correlated failures can be incorporated into a practical BFT system, several challenges remain. In particular, the approach relies on trustworthy information about replica attributes. Establishing properties such as geographic location, operational control, or the deployed software stack therefore requires suitable attestation mechanisms.

References

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